

109 p 40:

$$1) z^2 + 1 = 0 \Leftrightarrow z^2 = -1$$

$$\Leftrightarrow z = i \text{ ou } z = -i$$

$$S_1 = \{i; -i\}$$

$$2) z^2 + 2 = 0 \Leftrightarrow z^2 = -2$$

$$\Leftrightarrow z = \sqrt{2}i \text{ ou } z = -\sqrt{2}i$$

$$S_2 = \{\sqrt{2}i; -\sqrt{2}i\}$$

$$3) z^2 + 16 = 0 \Leftrightarrow z^2 = -16$$

$$\Leftrightarrow z = 4i \text{ ou } z = -4i$$

$$S_3 = \{4i; -4i\}$$

$$4) z^2 + 20 = 0 \Leftrightarrow z^2 = -20$$

$$\Leftrightarrow z = \sqrt{20}i \text{ ou } z = -\sqrt{20}i$$

$$\Leftrightarrow z = 2\sqrt{5}i \text{ ou } z = -2\sqrt{5}i$$

$$S_4 = \{2\sqrt{5}i; -2\sqrt{5}i\}$$

$$5) z^2 + \frac{1}{4} = 0 \Leftrightarrow z^2 = -\frac{1}{4}$$

$$\Leftrightarrow z = \frac{1}{2}i \text{ ou } z = -\frac{1}{2}i$$

$$S_5 = \{\frac{1}{2}i; -\frac{1}{2}i\}$$

$$6) z^2 + \frac{1}{3} = 0 \Leftrightarrow z^2 = -\frac{1}{3}$$

$$\Leftrightarrow z = \frac{1}{\sqrt{3}}i \text{ ou } z = -\frac{1}{\sqrt{3}}i$$

$$\Leftrightarrow z = \frac{\sqrt{3}}{3}i \text{ ou } z = -\frac{\sqrt{3}}{3}i$$

$$S_6 = \{\frac{\sqrt{3}}{3}i; -\frac{\sqrt{3}}{3}i\}$$

$$7) z^2 + \frac{11}{4} = 0 \Leftrightarrow z^2 = -\frac{11}{4}$$

$$\Leftrightarrow z = \frac{\sqrt{11}}{2}i \text{ ou } z = -\frac{\sqrt{11}}{2}i$$

$$S_7 = \{\frac{\sqrt{11}}{2}i; -\frac{\sqrt{11}}{2}i\}$$

$$8) z^2 + \frac{3}{2} = 0 \Leftrightarrow z^2 = -\frac{3}{2}$$

$$\Leftrightarrow z = \frac{\sqrt{3}}{\sqrt{2}}i \text{ ou } z = -\frac{\sqrt{3}}{\sqrt{2}}i$$

$$\Leftrightarrow z = \frac{\sqrt{6}}{2}i \text{ ou } z = -\frac{\sqrt{6}}{2}i$$

$$S_8 = \{\frac{\sqrt{6}}{2}i; -\frac{\sqrt{6}}{2}i\}$$